

Representation theorem for polyhedral cones

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We consider a polyhedral cone represented as a finitely generated set (note 1.(F) shows this is possible):

$$C := \text{cone}(E), \quad \text{where } E \subseteq \mathbb{R}^n \text{ is a non-empty finite set.}$$

The set of directions equivalent to $d \in C$ is denoted $R(d) := \{\alpha \cdot d : \alpha > 0\}$. If d is an extreme direction of C , then $R(d)$ is called an *extreme ray* of C .

In this note we prove the representation theorem for polyhedral cones: Every pointed polyhedral cone is the conic hull of its extreme direction:

Theorem 1 (*Representation theorem for pointed polyhedral cones*)

Suppose $C \neq \{0_n\}$, and that C is pointed. There exists a non-empty set $E^ \subseteq E$ satisfying:*

(i) $C = \text{cone}(E^*)$.

(ii) *The set $\{R(e) : e \in E^*\}$ is the unique set of extreme rays of C .* ■

Theorem 1 gives an essentially unique representation of a pointed polyhedral cone.

In the remainder we assume $C \neq \{0_n\}$, and that C is pointed. Every extreme direction of C is equivalent to a direction in E , which implies that the set of extreme rays of C is finite and unique:

Observation 1 (*Extreme directions are among the generators*)

The set of extreme rays of C is unique and a subset of $\{R(e) : e \in E\}$. ■

Indeed, any direction $d \in C \setminus \{0_n\}$ can be written $d = \sum_{e \in E} \mu_e \cdot e$ with $\mu \geq 0_{|E|}$. Therefore, if d is *not* equivalent to a direction in E , then d is by definition not extreme for C .

To prove Theorem 1.(i) we show non-extreme directions can be removed from E , *i.e.*, we show $C = \text{cone}(E \setminus \{e\})$ holds when $e \in E$ is not extreme for C . Applying this iteratively leaves a set $E^* \subseteq E$ that only contains extreme directions of C , *i.e.*, a set $E^* \subseteq E$ satisfying (i)-(ii) in Theorem 1.

Therefore let $\bar{e} \in E$ be an arbitrary non-extreme direction for C . In the remainder we show $C = \text{cone}(E \setminus \{\bar{e}\})$.

To show $C = \text{cone}(E \setminus \{\bar{e}\})$, it suffices to show $\bar{e}$ is conically generated by the directions in $E \setminus \{\bar{e}\}$, *i.e.*, it suffices to show $\bar{e} \in \text{cone}(E \setminus \{\bar{e}\})$:

Lemma 1 *If $\bar{e} \in \text{cone}(E \setminus \{\bar{e}\})$, then one has $C = \text{cone}(E \setminus \{\bar{e}\})$.* ■

Proof: " $\subseteq$ ": From $\bar{e} \in \text{cone}(E \setminus \{\bar{e}\})$ one obtains $E \subseteq \text{cone}(E \setminus \{\bar{e}\})$. Computing the conic hull on both sides of $E \subseteq \text{cone}(E \setminus \{\bar{e}\})$ gives:

$$C \subseteq \text{cone}(\text{cone}(E \setminus \{\bar{e}\})).$$

Since $\text{cone}(\text{cone}(S)) = \text{cone}(S)$ for every $S \subseteq \mathbb{R}^n$ (Corollary 1 in note 1.(A)), one has $\text{cone}(\text{cone}(E \setminus \{\bar{e}\})) = \text{cone}(E \setminus \{\bar{e}\})$. Hence $C \subseteq \text{cone}(E \setminus \{\bar{e}\})$.

" $\supseteq$ ": This inclusion follows directly by computing the conic hull on both sides of the inclusion $E \supseteq E \setminus \{\bar{e}\}$. ■

In the remainder we therefore show $\bar{e} \in \text{cone}(E \setminus \{\bar{e}\})$. Since $\bar{e}$ is not extreme for C , then $\bar{e}$ is a sum of directions in C non-equivalent to $\bar{e}$, *i.e.*, there exists directions $d^1, \dots, d^k \in C \setminus \{0_n\}$ such that $\bar{e} = d^1 + \dots + d^k$, where none of $d^1, \dots, d^k$ are equivalent to $\bar{e}$ (Observation 1 in note 1.(A)).

Let $i \in \{1, \dots, k\}$ be arbitrary. Since $d^i \in \text{cone}(E)$, the direction d^i is the sum of a direction in $\text{cone}(\{\bar{e}\})$, and a direction in $\text{cone}(E \setminus \{\bar{e}\})$, *i.e.*, one may write $d^i = \mu_i \cdot \bar{e} + y^i$ for some $\mu_i \geq 0$ and $y^i \in \text{cone}(E \setminus \{\bar{e}\})$. Since d^i is not equivalent to $\bar{e}$ one has $y^i \neq 0_n$. Summing all equations gives:

$$\bar{e} = d^1 + \dots + d^k = \mu^* \cdot \bar{e} + y^*,$$

where $\mu^* := \mu_1 + \dots + \mu_k \geq 0$ and $y^* := y^1 + \dots + y^k \in \text{cone}(E \setminus \{\bar{e}\})$.

We next argue that we have the following properties:

(a) $y^* \neq 0_n$:

If one had $y^* = y^1 + \dots + y^k = 0_n$, then $-y^1, y^1 \in C$, which implies C contains the line $\text{cone}(\{-y^1, y^1\}) = \text{span}(\{y^1\})$. This contradicts that C is pointed.

(b) $\mu^* < 1$:

Indeed, if $\mu^* \geq 1$, then $\bar{e} = \mu^* \cdot \bar{e} + y^*$ implies $-y^* = (\mu^* - 1) \cdot \bar{e} \in C$. However, then C contains the line $\text{cone}(\{-y^*, y^*\}) = \text{span}(\{y^*\})$. This again contradicts that C is pointed.

Hence $\mu^* \in [0, 1[$. The equation $\bar{e} = \mu^* \cdot \bar{e} + y^* \iff \bar{e} = \frac{1}{1-\mu^*} \cdot y^*$ with $y^* \in \text{cone}(E \setminus \{\bar{e}\})$ then shows $\bar{e} \in \text{cone}(E \setminus \{\bar{e}\})$. This completes the proof of Theorem 1.